Chapter 12

MCQ Practice Questions

Agent Factory Paradigm
Panaversity — General Agents Foundations

50 Practice MCQs

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MCQ Practice — Chapter 12

50	20	20	10
Total Questions	Easy	Medium	Hard

Instructions: Har question carefully padho. Answer check karne ke liye "Show Answer" button dabao. Print karte waqt answers automatically show honge.

Topics: Agent Maturity Model, LLM Constraints, OODA Loop, 5 Powers, AAIF Standards, SDD, Digital FTE, Enterprise Sales

Section 1: Agent Maturity Model & 2025 Inflection Point

Question 1 **Easy**

Agent Factory Paradigm mein "Incubator" ka doosra naam kya hai?

A) Custom Agent

B) General Agent

C) Specialist Agent

D) Digital FTE

Correct Answer: B) General Agent
Incubator = General Agent — explore ke liye use hota hai jab requirements clear na hon.

Question 2 **Easy**

Agent Maturity Model mein "Specialist" ko kya kehte hain?

A) General Agent

B) Incubator

C) Custom Agent

D) Foundation Model

Correct Answer: C) Custom Agent
Specialist = Custom Agent — ek specific kaam reliably karta hai.

Question 3 **Medium**

General Agents aur Custom Agents mein kya relationship hai?

A) Dono compete karte hain

B) General Agents Custom Agents ko replace karte hain

C) General Agents Custom Agents ko BUILD karte hain

D) Custom Agents General Agents se better hain

Correct Answer: C) General Agents Custom Agents ko BUILD karte hain
Yeh chapter ka sabse important concept hai — dono compete nahi karte!

Question 4 **Easy**

ICPC World Finals 2025 mein AI ne kitne problems solve kiye?

A) 8/12

B) 10/12

C) 11/12

D) 12/12 (Perfect Score)

Correct Answer: D) 12/12 (Perfect Score)

OpenAI ensemble ne perfect score kiya — sab 12 problems solve kiye, human teams se behtar.

Question 5 **Easy**

Stack Overflow 2025 Survey ke mutabiq kitne percent developers AI coding tools use karte hain?

A) 51%

B) 65%

C) 84%

D) 90%

Correct Answer: C) 84%
84% professional developers use ya plan karte hain AI tools use karna. 51% daily use karte hain.

Question 6 **Medium**

General Agent kab use karna chahiye?

A) Jab requirements crystallized aur clear hon

B) Jab workflow 1000+ baar chalana ho

C) Jab requirements unknown ya changing hon

D) Jab cost aur latency matter kare

Correct Answer: C) Jab requirements unknown ya changing hon
General Agent exploration ke liye hai — jab requirements clear na hon.

Question 7 **Medium**

"Premature Specialization" kaunsa anti-pattern hai?

A) General Agents production mein use karna

B) Requirements stabilize hone se pehle Custom Agent banana

C) Incubation skip karna

D) SDD use na karna

Correct Answer: B) Requirements stabilize hone se pehle Custom Agent banana
Premature Specialization = requirements clear hone se pehle Custom Agent banana — yeh galat hai.

Question 8 **Easy**

Digital FTE ki monthly cost roughly kitni hai?

A) $100-500/month

B) $500-2000/month

C) $4000-8000/month

D) $10,000+/month

Correct Answer: B) $500-2000/month
Digital FTE cost $500-2000/month hai jabke human FTE $4000-8000+/month.

Question 9 **Medium**

"Crystallized Understanding" se kya muraad hai?

A) Clear aur rigid specifications

B) Incubation phase se mila working code, patterns, aur insights

C) Custom Agent ka final output

D) Database schema design

Correct Answer: B) Incubation phase se mila working code, patterns, aur insights
Crystallized Understanding = incubation ka result — yeh "genetic material" ban jaata hai Specialist ke liye.

Question 10 **Hard**

Agent Factory evolution ka sahi sequence kya hai?

A) Specialist → Incubator → Production → Feedback

B) Planning → Coding → Testing → Deployment

C) Incubation → Crystallization → Specialization → Production → Improvement

D) Requirements → Design → Build → Test → Deploy

Correct Answer: C) Incubation → Crystallization → Specialization → Production → Improvement
Yeh complete evolutionary arc hai: Weeks 1-2 Incubation, Week 3 Crystallization, Weeks 4-5 Specialization, Week 6+ Production.

Section 2: LLMs ki 3 Core Constraints

Question 11 **Easy**

LLM ki pehli constraint kya hai?

A) Slow processing

B) Stateless — kuch yaad nahi rehta

C) Only English support

D) No internet access

Correct Answer: B) Stateless — kuch yaad nahi rehta
LLM har API call par naya start karta hai — koi memory nahi hoti previous conversations ki.

Question 12 **Easy**

Temperature = 0 ka matlab kya hai?

A) Most creative output

B) Highest randomness

C) Most deterministic (consistent) output

D) Model off hai

Correct Answer: C) Most deterministic (consistent) output
Temperature 0 = same input par same output. Temperature 1.0+ = high creativity but inconsistent.

Question 13 **Medium**

LLM ki "probabilistic" nature se kya problem aata hai?

A) Model bahut slow ho jaata hai

B) Same specs se alag-alag code aa sakta hai — validation mandatory hai

C) Model sirf English mein answer deta hai

D) Context window chota ho jaata hai

Correct Answer: B) Same specs se alag-alag code aa sakta hai — validation mandatory hai
Probabilistic nature ki wajah se TDD aur validation zaroori hai.

Question 14 **Easy**

Gemini 3 Pro ka context window kitna hai?

A) 200K tokens

B) 256K tokens

C) 1M tokens

D) 2M tokens

Correct Answer: D) 2M tokens
Gemini 3 Pro: 2M tokens (~5000 pages). Claude Opus 4.5: 200K. GPT-5.2: 256K.

Question 15 **Medium**

LLM ke "stateless" nature ka sahi mental model kya hai?

A) "AI mera colleague hai jo sab yaad rakhta hai"

B) "AI ek calculator hai"

C) "AI ek brilliant expert hai jise amnesia hai — har baar fresh briefing do"

D) "AI ek search engine hai"

Correct Answer: C) "AI ek brilliant expert hai jise amnesia hai — har baar fresh briefing do"
Yeh chapter ka exact quote hai — sahi mental model samajhna zaroori hai.

Question 16 **Medium**

Stateless constraint ka solution kya hai?

A) Har baar naya account banana

B) AGENTS.md, SPEC.md, PROJECT_CONTEXT.md files use karna

C) Model ko restart karna

D) Temperature badhaana

Correct Answer: B) AGENTS.md, SPEC.md, PROJECT_CONTEXT.md files use karna
Persistent files context ko preserve karte hain sessions ke across.

Section 3: OODA Loop & Coder to Orchestrator

Question 17 **Easy**

OODA Loop ka full form kya hai?

A) Output, Optimize, Decide, Adapt

B) Observe, Orient, Decide, Act

C) Operate, Order, Direct, Automate

D) Open, Organize, Deploy, Analyze

Correct Answer: B) Observe, Orient, Decide, Act
OODA = Observe, Orient, Decide, Act — yeh continuous cycle hai jisme autonomous agents kaam karte hain.

Question 18 **Easy**

Passive AI tools aur Agentic AI tools mein kya fark hai?

A) Passive tools expensive hain, Agentic tools saste hain

B) Passive tools predict karte hain, Agentic tools OODA cycle mein reason karte hain

C) Passive tools faster hain

D) Koi fark nahi

Correct Answer: B) Passive tools predict karte hain, Agentic tools OODA cycle mein reason karte hain
Agentic AI goal achieve karne tak OODA loop mein chalti rehti hai.

Question 19 **Medium**

AI ke saath kaam mein human ka sabse important contribution kya hai?

A) Code likhna

B) Boilerplate banana

C) Problem clarity, constraint awareness, aur quality standards (Judgment Layer)

D) Documentation likhna

Correct Answer: C) Problem clarity, constraint awareness, aur quality standards (Judgment Layer)
Human ka kaam Judgment Layer hai — AI execution karta hai, human judgment deta hai.

Question 20 **Medium**

Gen 4 AI tools mein (Claude Code, Gemini CLI) human ka role kya hai?

A) Typist

B) Prompt Engineer

C) Architect

D) Orchestrator

Correct Answer: D) Orchestrator
Gen 4 mein human Orchestrator ban gaya — "Definition of Done" define karta hai.

Question 21 **Hard**

Traditional vs AI-Orchestrated development mein ek feature ke liye kitna waqt lagta hai?

A) Traditional: 80 hours, AI: 80 hours (same)

B) Traditional: 140 hours, AI: 33 hours

C) Traditional: 100 hours, AI: 50 hours

D) Traditional: 200 hours, AI: 10 hours

Correct Answer: B) Traditional: 140 hours, AI: 33 hours
Planning same rehta hai lekin coding 80→8 hours, testing 30→3 hours, deployment 10→2 hours.

Section 4: Five Powers & Modern AI Stack

Question 22 **Easy**

AI Agents ki "5 Powers" kaunsi hain?

A) Speed, Accuracy, Memory, Cost, Scale

B) See, Hear, Reason, Act, Remember

C) Read, Write, Execute, Deploy, Monitor

D) Plan, Code, Test, Deploy, Monitor

Correct Answer: B) See, Hear, Reason, Act, Remember
5 Powers: SEE (visual), HEAR (audio), REASON (decision), ACT (execute), REMEMBER (context).

Question 23 **Easy**

Modern AI Stack ki Layer 1 kya hai?

A) Agent Skills

B) AI-First IDEs

C) Frontier Models (Claude, GPT, Gemini)

D) MCP Servers

Correct Answer: C) Frontier Models (Claude, GPT, Gemini)
Layer 1: Frontier Models (Brain), Layer 2: AI-First IDEs (Context), Layer 3: Agent Skills (Worker).

Question 24 **Medium**

AI Evolution ke 3 phases sahi order mein kya hain?

A) Generative → Predictive → Agentic

B) Agentic → Predictive → Generative

C) Predictive → Generative → Agentic

D) Predictive → Agentic → Generative

Correct Answer: C) Predictive → Generative → Agentic
Pehle predict karna seekha, phir create karna, phir autonomous action lena.

Question 25 **Medium**

MCP aur Agent Skills mein kya fark hai?

A) Dono same kaam karte hain

B) MCP = connectivity (haath), Skills = domain expertise (training)

C) MCP = training, Skills = connectivity

D) MCP sirf Anthropic ke liye hai

Correct Answer: B) MCP = connectivity (haath), Skills = domain expertise (training)
MCP = Agent ke haath (tools se jodta hai), Skills = Agent ki training (kaise kaam karna hai).

Section 5: AAIF Standards

Question 26 **Easy**

AAIF kab announce hua?

A) January 2025

B) June 2025

C) December 9, 2025

D) March 2026

Correct Answer: C) December 9, 2025
AAIF (Agentic AI Foundation) December 9, 2025 ko Linux Foundation ke under announce hua.

Question 27 **Easy**

MCP kahan se aaya (originally)?

A) OpenAI

B) Google

C) Anthropic

D) Microsoft

Correct Answer: C) Anthropic
MCP November 2024 mein Anthropic ne open source release kiya. March 2025 mein OpenAI ne adopt kiya.

Question 28 **Medium**

MCP ke 3 Universal Primitives kaunse hain?

A) Models, Agents, Tools

B) Resources, Tools, Prompts

C) Input, Process, Output

D) Read, Write, Execute

Correct Answer: B) Resources, Tools, Prompts
Resources (read-only data), Tools (state-changing actions), Prompts (reusable templates).

Question 29 **Medium**

AGENTS.md kya karta hai?

A) Agents ko internet access deta hai

B) Agent ko local project ke rules sikhata hai

C) Agents ka password store karta hai

D) Project documentation hai

Correct Answer: B) Agent ko local project ke rules sikhata hai
AGENTS.md = "Yahan kaise behave karna hai" — code style, security rules, architecture patterns.

Question 30 **Hard**

Progressive Disclosure mein Level 1 par kitne tokens load hote hain?

A) 50 tokens

B) ~100 tokens

C) 5,000 tokens

D) 10,000 tokens

Correct Answer: B) ~100 tokens
Level 1: ~100 tokens (naam aur description). Level 2: 5K tokens (full SKILL.md). Result: 80-98% token reduction.

Question 31 **Medium**

goose kahan se aaya?

A) Anthropic

B) Google

C) Block

D) Microsoft

Correct Answer: C) Block
goose Block ki taraf se hai — local-first execution, MCP-native design. 75% Block engineers 8-10+ hours weekly save karte hain.

Section 6: Digital FTE Business Strategy

Question 32 **Easy**

Human customer support agent per ticket cost kitni hai vs Digital FTE?

A) Human: $50, Digital FTE: $10

B) Human: $150, Digital FTE: $3

C) Human: $100, Digital FTE: $20

D) Human: $200, Digital FTE: $50

Correct Answer: B) Human: $150, Digital FTE: $3
Human: $6000/month, 20 tickets/day = $150/ticket. Digital FTE: $1500/month, 500+ tickets/day = $3/ticket. 50x efficient!

Question 33 **Medium**

"90/10 Split" mein "10% Moat" kya hai?

A) 10% code jo AI nahi likh sakta

B) Domain expert ka nuance, edge cases, aur experience-based judgment

C) Business ka 10% profit margin

D) Prompt engineering skills

Correct Answer: B) Domain expert ka nuance, edge cases, aur experience-based judgment
Prompt engineering moat nahi — domain expertise moat hai. AI kya galat kar sakta hai, sirf expert jaanta hai.

Question 34 **Medium**

License model (AAIF standard mein) kis ke liye best hai?

A) Small businesses aur startups

B) Skeptical markets

C) Enterprise aur regulated industries (data on-premise)

D) Volume plays aur passive income

Correct Answer: C) Enterprise aur regulated industries (data on-premise)
License model: $5,000-50,000/year. Client ke server par install hota hai — compliance advantage.

Question 35 **Hard**

PPP Framework (Piggyback Protocol Pivot) mein Phase 1 kya hai?

A) Independent platform banana

B) Market validation via vendor marketplaces

C) Infrastructure Layering — MCP aur Agent Skills standardize karna

D) Enterprise sales karna

Correct Answer: C) Infrastructure Layering — MCP aur Agent Skills standardize karna
Phase 1 (Months 0-6): Infrastructure Layering. Phase 2 (Months 6-18): Market Validation. Phase 3 (18+): Independent Platform.

Section 7: SDD & Nine Pillars of AIDD

Question 36 **Easy**

SDD ke 6 phases sahi order mein kya hain?

A) Plan → Code → Test → Deploy → Review → Fix

B) Specify → Clarify → Plan → Tasks → Implement → Validate

C) Research → Design → Build → Test → Launch → Monitor

D) Ideate → Prototype → Test → Iterate → Scale → Maintain

Correct Answer: B) Specify → Clarify → Plan → Tasks → Implement → Validate
Yeh SDD ka exact 6-phase workflow hai — exam mein zaroor aayega!

Question 37 **Easy**

SDD mein ek task kitne time se zyada nahi hona chahiye?

A) 30 minutes

B) 1 hour

C) 2 hours

D) 4 hours

Correct Answer: C) 2 hours
Tasks phase mein: "No single task exceeds 2 hours" — chote manageable pieces.

Question 38 **Medium**

"Vibe Coding" kab theek hai?

A) Production features ke liye

B) Security-critical work ke liye

C) Learning experiments aur throwaway prototypes ke liye

D) Team projects ke liye

Correct Answer: C) Learning experiments aur throwaway prototypes ke liye
Vibe Coding = bina spec ke — sirf experiments mein theek hai, production mein dangerous.

Question 39 **Medium**

AIDD mein "Markdown as Programming Language" ka matlab kya hai?

A) Markdown se apps build hoti hain

B) Markdown-formatted specs AI agents ke liye executable "source code" ban jaati hain

C) Markdown replace karta hai Python ko

D) Documentation format hai

Correct Answer: B) Markdown-formatted specs AI agents ke liye executable "source code" ban jaati hain
Pillar 2: Markdown specifications agent ke liye instructions hain — source of truth.

Question 40 **Hard**

AI ke saath "Vague requirements" ka effect kya hota hai?

A) Mediocre results, slow

B) Good results, just slower

C) Terrible results, FAST (multiplied)

D) Same as without AI

Correct Answer: C) Terrible results, FAST (multiplied)
AI discipline ko amplify karta hai — vague requirements ke saath AI bura result aur bhi tezi se deta hai!

Question 41 **Medium**

M-Shaped Developer kya hota hai?

A) Sirf ek field mein expert

B) Ek deep, baaki shallow

C) 2-4 complementary domains mein deep expertise (AI ki wajah se possible)

D) Sabhi fields mein shallow knowledge

Correct Answer: C) 2-4 complementary domains mein deep expertise (AI ki wajah se possible)
M-Shaped = new developer profile. Pehle impossible tha, ab AI augmentation se possible hai.

Section 8: Enterprise Sales & Digital FTE Vision

Question 42 **Easy**

Agentic AI services ka McKinsey estimate market size kya hai?

A) $10-50 Billion

B) $50-100 Billion

C) $100-400 Billion

D) $1 Trillion+

Correct Answer: C) $100-400 Billion
McKinsey estimate: $100-400 Billion in business function transformation. Top 5 industries focused.

Question 43 **Medium**

CFO ko kya boltay hain agentic AI sell karte waqt?

A) "Hamara AI sabse accurate hai"

B) "12 mahine mein ROI milega"

C) "50% resolution time kam hoga"

D) "Built-in security aur governance hai"

Correct Answer: B) "12 mahine mein ROI milega"
CFO = financial ROI. COO = operations efficiency. CIO/CTO = security/technical. CEO = competitive advantage.

Question 44 **Medium**

"Gen AI Paradox" kya hai?

A) Gen AI bahut expensive hai

B) 75%+ organizations Gen AI use karte hain lekin material earnings impact nahi

C) Gen AI sirf enterprises ke liye hai

D) Gen AI security risk hai

Correct Answer: B) 75%+ organizations Gen AI use karte hain lekin material earnings impact nahi
Yahi frustration opportunity hai — agentic AI measurable results deta hai jo traditional Gen AI nahi deta.

Question 45 **Hard**

Enterprise implementation ki 5-phase roadmap mein Phase 1 kya hai?

A) Scaled Deployment

B) Rapid Prototyping

C) Discovery and Strategy (4-6 weeks)

D) Pilot Deployment

Correct Answer: C) Discovery and Strategy (4-6 weeks)
Phase 1: Discovery (4-6 weeks) → Phase 2: Prototyping (6-8 weeks) → Phase 3: Pilot (8-12 weeks) → Phase 4: Scale → Phase 5: Operations.

Section 9: Mixed Revision Questions

Question 46 **Hard**

Kaunsa anti-pattern hai jab aap sirf General Agent se hi production chalate rehte ho?

A) Premature Specialization

B) Perpetual Incubation

C) Skipping Incubation

D) Over-Specification

Correct Answer: B) Perpetual Incubation
Perpetual Incubation = production workloads ke liye General Agents use karte rehna — yeh galat hai.

Question 47 **Medium**

Shadow Mode Deployment Strategy mein Week 1-4 mein kya hota hai?

A) Agent fully autonomous hai

B) Agent recommend karta hai, human final decision leta hai

C) Human review karta hai only high-risk cases

D) Agent aur human mil ke decide karte hain

Correct Answer: B) Agent recommend karta hai, human final decision leta hai
Phase 1 (1-4 weeks): Agent generates recommendations; humans make ALL final decisions.

Question 48 **Medium**

SDD ka core premise kya hai?

A) Tez code likhna zaroori hai

B) Bottleneck ab implementation speed nahi — clarity of requirements hai

C) AI ko zyada freedom deni chahiye

D) Testing optional hai

Correct Answer: B) Bottleneck ab implementation speed nahi — clarity of requirements hai
Yahi SDD ka central premise hai — AI ke saath speed bottleneck nahi, clarity bottleneck hai.

Question 49 **Hard**

AIDD mein "Composable Vertical Skills" se kya muraad hai?

A) Vertical line mein code likhna

B) Reusable domain expertise modules jo agents specific problems ke liye load kar sakein

C) Vertical market mein sales karna

D) Database schemas

Correct Answer: B) Reusable domain expertise modules jo agents specific problems ke liye load kar sakein
Pillar 8: Composable Vertical Skills = portable, reusable domain expertise packages (SKILL.md).

Question 50 **Hard**

Chapter 12 ka "final binary choice" kya hai?

A) Claude use karo ya GPT use karo

B) Custom Agent banao ya General Agent use karo

C) Path A: AI = accelerated typing (vibe coding), ya Path B: Agent Factory master karo (SDD +

Digital FTEs)

D) Enterprise mein ja ya startup banao

Correct Answer: C) Path A: AI = accelerated typing, ya Path B: Agent Factory master karo
Yeh chapter ka closing message hai: "The question isn't whether AI will change how you build — it's whether you'll lead or follow."

Mubarak Ho! 🎉

Aapne Chapter 12 ke saare 50 MCQs complete kar liye!

Ab Chapter 13 aur 14 ki taraf badho.

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